Assignment 2: Data wrangling

EVR 628- Intro to Environmental Data Science

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The big picture

Remember that the final goal is to have a GitHub repository where you can showcase your work. Assignment 1 was to create the repository, which should be essentially empty. Assignment 2 is to **develop two R scripts to clean and explore a dataset of your choice**. Your choice of data will lock you in a path: The third assignment will be to formally visualize the data you have cleaned up here. Your final project will leverage the data and visualizations you'll produce to wrap it all together.

This assignment

Tasks: Develop two R scripts. One will be a data wrangling script that reads raw data (*e.g.* in `.xlsx` or `.csv`), performs data cleaning, transformation, or wrangling as needed, and exports a processed version of the data (`.rds`) to be used in later assignments. The second script read in the clean data and performs exploratory data visualizations, which will be exported as `*.png` files.

This assignment will require quite a bit of work on three fronts:

1. Coming up with an idea for a final project
2. Thinking about how you will get the raw data into a format that gets you closer to your final project, and
3. Using lecture and live coding session materials–and function documentation–to help you build your cleaning and visualization pipeline.

I have included a list of potential data sources and ideas for final projects. Feel free to come up with your own ideas. I recommend choosing something you may be genuinely interested in, which will keep you motivated.

Grading Rubric (and checklist!)

- **10%**: `README.md` file, split as:
  - ☐ 5% Explains the main objective of your project in 2-3 sentences.

    - ☐ 5% Lists the contents of the project repository:
    - ☐ 2.5%: The names and sources of raw data are clearly listed
    - ☐ 2.5%: The names and location of R scripts, clean data, and figures are clearly listed
- **40%**: Your repository contains an R script called `data_processing.R`, saved to your `scripts/01_processing` folder that:
    - ☐ 5%: Clearly indicates packages loaded at the top of the script
    - ☐ 5%: Uses **relative** paths to read AND write files
    - ☐ 5%: Contains code documentation using comments via `#`
    - ☐ **20%: Uses `dplyr` verbs and / or `tidyr` functions to achieve the data cleaning task.**
    - ☐ 5%: The exported data conform to `tidy` data standards regarding rows, columns, and values.
- **40%**: Your repository contains a script called `exploratory_data_visualization.R`, saved to your `scripts/03_content` folder that:
    - ☐ 5%: Clearly indicates packages loaded at the top of the script
    - ☐ 5%: Uses **relative** paths to read AND write files
    - ☐ 5%: Contains code documentation using comments via `#`
    - ☐ **20%: Uses `ggplot2` functions to explore your data in at least three plots.**
    - ☐ 5%: The exported plots have:
        - ☐ axis labels
        - ☐ a title
        - ☐ a caption indicating source of data
- **10%** : Reproducibility. I can clone your repo and reproduce your clean data and figures without having to modify your code.

> **!** Important points
>
> **Don't be lazy -** Simply renaming columns or changing their order is not enough. Come talk to me, I guarantee we can find something to do to them.

Some examples of repositories that would get a 100%:

- Analyzing mid-term survey data
- Building a geospatial dataset of Mexican ports, by McGill undergraduate student Emma Zgonena

Turning in your assignment

- Please share the link to your github repo via Canvas
- The deadline for this assignment is October 11, 2026 by 11:59

Project ideas with different data sets

Below are examples of data sources and final project ideas. For the purpose of this assignment you are not expected to provide an answer. You will simply prepare the data necessary to eventually provide the answer.

Tuna Data

Access tuna data from the WCPFC or IATTC and then:

- Build a time series of catch-per-unit-effort for a species of your liking
- Find the year in which total tuna catch peaked
- Find the month where tuna catch is, on average, maximum
- For every year, find the coordinates where tuna catch was the highest. Do we see any changes in it's location?

Oceanic Indices

Go to NOAA's Physical Sciences Laboratory and find an interesting data set

- Use monthly NINO3 to produce a figure of NINO / NINA years
- Compare indices among them
- Relate indices here to tuna catch data above

STRAVA data

Export your data from strava[1], and then:

- Find your longest activity
- Find your total traveled distance since you started logging activities
- Find the average length of your runs
- Analyze your trends in heart rate vs speed
- Analyze the pace of runs performed with the Rosenstiel Running Club

Dive watch data

Your dive watch should allow you to export your data[2], although you might need a special cable / software.

- Calculate how much time you've spent underwater per week, month, or year
- If you have air pressure info associated (or you keep a dive log with it), calculate your Surface Air Consumption Rate
- What's your longest dive?
- What's your deepest dive?
- What is the average depth of all your dives?

[1] Data will not exported as `.csv` but I can help work around that
[2] The data may not exported as `.csv` but I can help work around that

- What is the maximum number of dives you have done in a single day? a week?

Data from papers

The following papers have data available. You could download their data and recreate or augment their figures.

- Burguess et al, 2018. Protecting marine mammals, turtles, and birds by rebuilding global fisheries | GitHub repo
- Kuczenski et al. 2021Plastic gear loss estimates from remote observation of industrial fishing activity | GitHub repo
- Clark et al., 2023 Global assessment of marine plastic exposure risk for oceanic birds | GitHub repo
- A database of mapped global fishing activity 1950–2017
- O'Connor et al., 2024 Effects of anthropogenic noise on marine mammal abundances informed by mixed methods
- Jouffray et al., 2025. Identifying and closing gaps in corporate reporting of ocean impacts
- Oyanedel et al., 2025. Improving detectability of illegal fishing activities across supply chains | GitHub repo

Other sources of data

- The National Center for Environmental Information NCEI contains loads of interesting data sets.
- Animal tracking data from MoveBank
- Vessel tracking data from Global Fishing Watch

Synthetic data

- Maybe you are in the midst of collecting data for your project. You don't yet have the raw data, but you have a sense of what they will look like (number of columns, number of observations…). I can help you simulate the data and you can build a script that helps you get your data into an analysis-ready format.

None of these are working for you?

Come to office hours and we can discuss ideas.